CHAPTER 9

exponential function and models continuous

Chapter overview

Objectives. Use algebraic properties; drift; study growth or decline ; connect exponential and geometric sequences.
Prerequisites. Power, geometric sequences, bypass and sign.
Annual benchmark. 2·11 February: 8 a.m.

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Simplify 2 $^{3} \times 2^{4}$.

2. Express 3 $^{-2}$.

3. Resolve 2 $x + 1 = 5$.

4. Derivating $x^{2} + 2x$.

5. A decrease of 20% gives what coefficient?

6. For a geometric sequence $u_0 = 5, q = 2$, calculate u_n.

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Explain the difference between continuous rate and discrete multiplier coefficient; present an inequality proved by minimum difference.

Search before formalizing

Compare the increments of e^x and the values of the function around 0, with several steps
Build three stages of Euler and explain the chosen rule.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

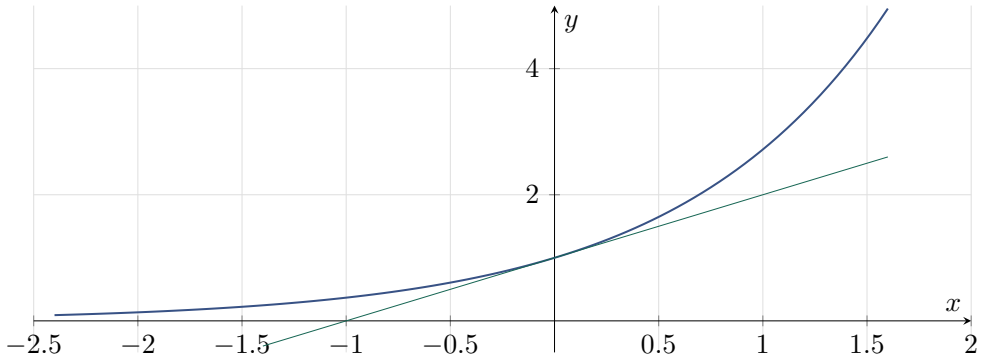
9.1 Defining a new function

Definition and results

The exponential function is the only ferivable function on $\mathbb{R}$ verifier $f\prime = f$ and $f(0) = 1$.
Its existence and uniqueness are admitted. expthen e^x with $e = \exp(1) \approx 2{,}71828$.
The number e is a constant; x is the variable. The exponential is strictly positive.

Status and justification. Existence and uniqueness admitted in accordance with the programme.
Deduced, after the algebraic properties allowed below, from $\exp(x) = \exp(\frac{x}{2})^2$ and $\exp(x)\exp(-x) = 1$.

Example solved. $\exp(0) = 1$, $\exp(1) = e$, exp ′(0) = 1. The tangent in 0 is therefore $y = 1 + x$. The Rating e^x extends power properties to any real exhibitor.



e^x (blue) and its tangent 1 + x in 0 (green).

Point of vigilance

e^0 = 1, not 0. The function never takes the value 0, even for negative xes.

Immediate application. What is the equation of tangent to e^x One?

9.2 Algebraic properties and comparisons

Definition and results

For all real x,y, $e^{x+y} = e^x e^y$, $e^{-x} = \frac{1}{e^x}$, $e^{x-y} = \frac{e^x}{e^y}$. For integer n, ($e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, therefore $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Status and justification. The relationship of addition is accepted in the common path; it can be deepened through uniqueness. Positivity and $\exp' = \exp > 0$ give the strict growth. justifies equivalences on equations and inequities.

Example solved. $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ equals 3 $x - 2 = x$ +4, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Point of vigilance

e^{x+y} is a product, never in general $e^x + e^y$. We don't solve e^x = 2 with logarithm Not studied first.

Immediate application. Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

9.3 Derive and Study

Definition and results

For real, $t \mapsto e^{at}$ is derivable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$the derivative is ae^{at+b}. Product rules and quotients apply with already known derivatives.
It often simplifies the study of the derivative sign.

Status and justification. The formula of derivation of e^{at} is allowed; the case with b ensues from it exiting the constant factor e^b. The general compound bypass is not necessary here.

Example solved. For $f(x) = (x+1)e^{-x}$, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. Like $e^{-x} > 0$, the derivative has the sign of $-x$: total maximum 1 in 0.

Point of vigilance

Do not forget the factor has nor the negative sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Immediate application. Dirty $g(t) = 3e^{-2t}$ and study its sense of variation.

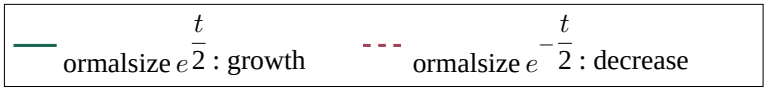
9.4. From discrete to continuous

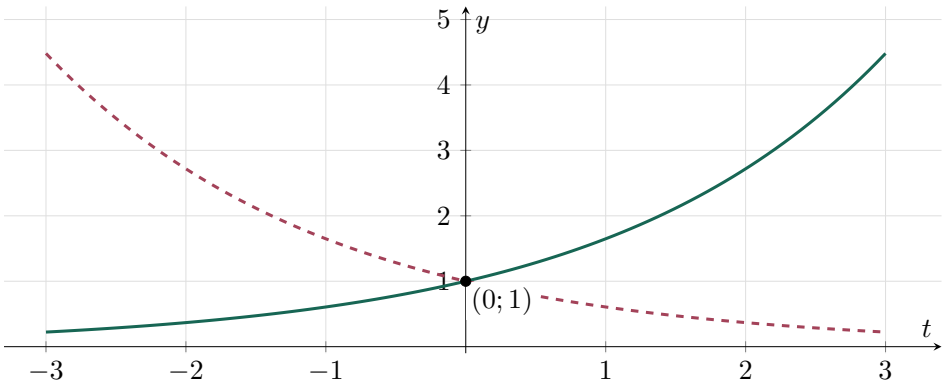
Definition and results

A model $Q(t) = Q_0 e^{kt}$, $Q_0 > 0$, represents growth if $k > 0$, a decrease if $k < 0$, a consistency if k = 0. For a fixed pitch h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. Values at moments integer form a geometric sequence of common ratio e^k. The unit of k is the inverse of the unit of time.

Status and justification. The report is based on: $e^{k(t+h)} = e^{kt}e^{kh}$. The direction of variation is derived from $Q\prime(t) = kQ_0e^{kt}$ and positivity of the remaining factor.

Example solved. $Q(t) = 100e^{-0{,}2t}$T in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0{,}2}$. A simulation can look for the first integer number of hours where Q passes under 50; the exact equation is not solved by logarithm in this chapter.





The curves corresponding to $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetrical in relation to the vertical axis. trait styles also distinguish them without color.

Point of vigilance

The relative rate per unit of discrete time is $e^k - 1$, not exactly k; they are close Only for k small.

Immediate application. For $Q(t) = 5e^{0{,}3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

9.5 Digitally Build and Demonstrate Inequality

Definition and results

To approach the solution $y' = y,\ y(0) = 1$ with one step h, Euler replaces the increase by $hy : y_{k+1} = (1+h)y_k$. On [0; 1] with $h = \frac{1}{n}$, the approximation of e is $\left(1+\frac{1}{n}\right)^n$. This construction is experimental. $e^x \geq 1 + x$ obtained by function study.

Status and justification. For $g(x) = e^x - 1 - x,\ g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves inequality; equality only in 0. Convergence o Euler is not demonstrated here.

Example solved. With n=2, Euler gives $y_0 = 1, y_1 = 1{,}5, y_2 = 2{,}25$. With n=10, $(1\ ,1)^{10} \approx 2{,}59374$. We compare to e; we do not confuse approximate value with function definition.

Point of vigilance

A graph showing a curve above a straight line does not prove inequality over all the real.

Immediate application. Show that $e^{2x} \geq 1 + 2x$ for all real x.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance. to 7 are a reserve of transfer, research or programming: the professor designates one. Exercise 8 resumes at a distance.

Exercise 9.1 · Transform

For all real x, we consider the expressions

$$A = e^x e^{2-x}, \qquad B = \frac{(e^{2x})^3}{e^x}, \qquad C = e^{-x}(e^x + 1).$$

1. Simplify A and B justifying each grouping of exhibitors.

2. Develop and simplify C. Explain why a sum does not simplify in the same way a product.

Exercise 9.2 · Equations and inequities

An equation and a real unknown inequitable is considered x :

$$e^{2x-1} = e^{x+3} \quad \text{and} \quad e^{-3x} \geq e^{x-2}.$$

1. Resolve each in $\mathbb{R}$, citing the property of the exponential which allows comparison Exhibitors.

2. When solving the inequitable, explain at what stage the sense of inequality changes. all solutions in the form of interval.

Exercise 9.3 · Product and maximum

The function f is defined on [0; 4] by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study his sign over the interval.

2. Draw up the table of changes in f. Deduce its maximum and the abscisse where it is reached.

3. Check that the values at the bounds do not give a greater maximum.
exact values.

Exercise 9.4 · A modeled population

The following two parts are independent.

Part A · A population. A population is modelled by $P(t) = 200e^{0,1t}$where: $t \geq 0$ is the time in years. $P(t)$ is a theoretical workforce, which is not necessarily integer.

1. Determine initial strength, then multiplier coefficient between moments t and $t + 1$. In deduct the exact annual rate and its percentage rounded to one hundredth.

2. Explain why this model cannot necessarily describe a real population indefinitely.

3. **Part B · Two curves.** In the same coordinate system, outline the curves of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$.
Specify their variations, common point and symmetry. Clearly identify each curve.

Exercise 9.5 · Search for a threshold

The quantity of a substance, in milligrams, is modelled at the moment $t \geq 0$ per $Q(t) = 80e^{-\frac{t}{2}}$, where t is expressed in hours. Checks take place at full moments $n = 0, 1, 2, \ldots$. We're looking for the for which the quantity is strictly less than 10 mg.

1. Write a search algorithm of this first integer. Provide a bounder and specify
what the program returns if the bound is reached.

2. Run the algorithm, then numerically check the amount in the row found and in the previous row.

3. To justify, with the help of the decrease of Q, that this double check establishes that it is indeed the
do not use logarithm.

Exercise 9.6 · A symmetry

The function is considered f defined on $\mathbb{R}$ by $f(x) = e^x + e^{-x}$.

1. Studying gender parity f and interpret the result for its curve.

2. Demonstrate that $f(x) \geq 2$ for all real x, by choosing a method. Specify the case of equality.

3. Propose a second method of checking the result, from a square or from a study of
changes.

Exercise 9.7 · Singleness under a hypothesis

Deepening. It is assumed that the exponential function has already been built. f a derivable function on $\mathbb{R}$ who verifies $f'(x) = f(x)$ for all real x and $f(0) = 1$. $g(x) = f(x)e^{-x}$.

1. Calculate g' and explain why g is constant on $\mathbb{R}$.

2. Determine this constant and then deduce from it the expression of f.

3. Does this demonstration establish the existence or uniqueness of the function sought? Justify.

Fiscal year 9.8 · Deferred review

The sequence is defined (u_n) by $u_n = e^{2n+1}$ for all natural n.

1. Demonstrate that this sequence is geometric, then specify its first term and common ratio.

2. Calculate $u_0 + u_1 + u_2$ In exact form. Compare the direct calculation and use of a formula of the sum.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 187, questions a, b and c. Simplify products and quotients by quoting the ownership of the exponentials employed.

CQFD, exercise 30 p. 187, questions a, b and c. Solve the three inequities and write the Sets of solutions in the form of intervals.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $e^0 =$

   A. 0
   B. 1
   C. e
   D. -1

2. $e^x e^y =$

   A. e^{x+y}
   B. e^{xy}
   C. $e^x + e^y$
   D. e^{x-y}

3. Derivative from e^{-2x} :

   A. e^{-2x}
   B. $2e^{-2x}$
   C. $-e^{-2x}$
   D. $-2e^{-2x}$

4. e^x = 1 equals

   A. $x = 1$
   B. $x = e$
   C. $x = 0$
   D. No solution

5. The sign of e^{-x} is

   A. negative if x>0
   B. Always strictly positive
   C. None in 0
   D. Variable

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $e^{2x} < e^3$ is equivalent to

   A. $x < 3$
   B. $x > \frac{3}{2}$
   C. $x < \frac{3}{2}$
   D. $x < \sqrt{3}$

2. $u_n = e^{3n}$ a for reasons

   A. e^3
   B. 3

C. e

D. $3e$

3. Tangente to e^x in 0:

A. $y = x$

B. $y = ex$

C. $y = 1$

D. $y = x + 1$

4. Derivative from xe^x :

A. e^x

B. $(x + 1)e^x$

C. xe^x

D. xe^{x-1}

5. $e^x \geq 1 + x$ Equality

A. For all x

B. in 1 only

C. in 0 only

D. Never

True or false?

1. $e^{2x} = 2e^x$ for all x.

2. The function $t \mapsto e^{-3t}$ is decreasing.

3. A continuous rate k=0.1 corresponds exactly to 10% increase in one unit of time.

Personal review

I keep the rules of the powers; I know the positivity; I drift with the factor; I separate
model, exact calculation and simulation.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and
a mistake to avoid.

Corrected and conduct of chapter 9

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $2^7 = 128$. *Action:* Reconstitute the product by seven factors.
2. $\frac{1}{9}$. *Action:* Back to the reverse power.
3. $x = 2$. *Action:* Isolate the unknown while maintaining equivalence.
4. $2x + 2$. *Action:* Apply the sum of the derivatives.
5. $0{,}8$. *Action:* Subtract the rate to 1.
6. 5×2^n. *Action:* Count the n multiplications.

Research activity

The condition $f\prime = f$ Motivation$y \approx hy$. With h= $\frac{1}{4}$, the values are 1 ; 1.25 ; 1.5625 ; 1.953125.
construction approximates the function and does not replace its characterization.

Immediate applications

1. $y = e + e(x - 1) = ex$.
2. e^{2x+1} ; $x + 2 > 2x$, therefore $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: Strictly decreasing.
4. $u_n = 5(e^{0{,}3})^n$; the report on two units is $e^{0{,}6}$.
5. Apply demonstrated inequality with argument 2 x ; equality only in x=0.

Exercises: approach and results

Exercise 9.1 · Transform. e^2 ; e^{5x} ; $1 + e^{-x}$. For the last expression, distribute and simplify products.

Exercise 9.2 · Equations and inequities. By strict growth: 2 $x - 1 = x$ + 3, i.e. x=4.
Then $-3x \geq x - 2$, or $x \leq \frac{1}{2}$.

Exercise 9.3 · Product and maximum. $f\prime = (1 - x)e^{-x}$; growth up to 1 and then decline.
Maximum $f(1) = \frac{1}{e}$. Values at bounds 0 and 4 e^{-4} lower.

Exercise 9.4 · A modeled population. 200 ; factor $e^{0{,}1}$; rate $e^{0{,}1} - 1 \approx 10{,}52$ per cent
resources and the capacity of the environment are not taken into account; it is a model over a period of time
The two curves requested go through (0;1); the first one grows, the second one decreases.
are symmetrical in relation to the vertical axis; see the figure in the section From Discrete to Continuous.

Exercise 9.5 · Search for a threshold. Test successively n=0,... with a limit of iterations. $Q(4) \approx 10{,}8268 \geq 10$, $Q(5) \approx 6{,}5668 < 10$; Demonstrated decay, therefore first integer 5.

Exercise 9.6 · Symmetry. $f(-x) = f(x)$: pair. $f' = e^x - e^{-x}$ has the sign of x, so minimum 2 in 0. Other evidence: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^2$.

Exercise 9.7 · Singleness under one hypothesis. $g' = f'e^{-x} - fe^{-x} = 0$. $\mathbb{R}$, g is constant, and g(0)=1. $f(x) = e^x$. This proof of uniqueness uses an exponential already constructed, it does not does not demonstrate existence.

Exercise 9.8 · Deferred review. $\frac{u_{n+1}}{u_n} = e^2$; first term e. Sum $e(1+e^2+e^4) = \frac{e(1-e^6)}{1-e^2}$. The direct sum is also effective here.

CQFD, exercise 18 p. 187, questions a, b and c. a : e^{3x}. b : e^2, since e^{x+2} is divided by e^x, Always non-zero. c: e^{x+1}. Do not confuse the number e with the variable x.

CQFD, exercise 30 p. 187, questions a, b and c. a : $x \leq 0$. b : $2x > 1$, therefore $x > \frac{1}{2}$. c : $-3x \geq 2x + 1$, therefore $x \leq -\frac{1}{5}$. Strict growth of the exponential keeps meaning; division by -5 reverses it in c.

MCQ: answers and reasons

MCQ 1.

1. B. Initial condition.
2. A. Property of addition.
3. D. Factor -2.
4. Strict growth and $e^0 = 1$.
5. B. Positiveness of the exponential.

MCQ 2.

1. C. Compare exhibitors.
2. A. Report of consecutive terms.
3. D. Value 1 and slope 1.
4. B. Product rule.
5. C. Minimum difference.

True or false

1. False: in 0, 1 is different from 2.
2. True: its derivative $-3e^{-3t}$ is negative.
3. False: the discreet rate is valid $e^{0,1} - 1$, about 10.52 per cent.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.